VISHAQ JAYAKUMAR

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EDUCATION

Northeastern University, Boston *Jan 2023 - May 2025*
Master of Science in Computer Science *GPA: 3.7*

Sri Sivasubramaniya Nadar College of Engineering, Chennai *Aug 2018 - Apr 2022*
B.Tech, Information Technology *GPA: 3.7*

WORK EXPERIENCE

Aura Intelligent Systems, Boston Sept 2024 – Dec 2024
Machine Learning Intern

- Streamlined **nuScenes** dataset access using **Python** APIs, improving model training speed for perception systems.
- Reduced real-time object-detection latency 25% by optimizing a lightweight detector for **AMD Versal AI Engine (Python, C++)**.
- Converted raw **TI AWR1243** radar data to point clouds using **Python and MATLAB** for perception and sensor fusion.
- Developed radar preprocessing filters and clustering algorithms to enhance **ADAS** object detection.
- Improved object tracking reliability by **30%** by integrating radar and vision-based **clustering** outputs, significantly reducing false positives in autonomous navigation.
- Implemented radar–vision synchronization; reduced skew and jitter, boosting multi-sensor tracking reliability.

LARSEN & TOUBRO (L&T), Chennai Apr 2021 – May 2021
Software Engineering Intern

- Built a responsive **Angular + C#** ops dashboard with live charts and RBAC; improved realtime visibility by 30%.
- Designed **RESTful APIs** and service integrations; deployed on **AWS (EC2, RDS)** via **CodeDeploy** with **S3/IAM** and env configs.
- Automated **CI/CD** with **GitHub Actions**; added tests, health checks, and rollback; cut deploy time 60% and failures 35%.
- Implemented **CloudWatch** logs, metrics, and alarms for release health; reduced mean time to detect and recover issues.

Verzeo, Bangalore Jan 2020 - Feb 2020
Machine Learning Intern

- Built a non-invasive **MGMT methylation** predictor from imaging features with **normalization**, **augmentation**, and clean splits.
- Re-implemented the classifier in **PyTorch** with transfer learning; achieved **85%** under **5-fold CV** with **stratified** splits.
- Accelerated training with **Distributed Data Parallel** on **GPUs** and **mixed precision**; tuned **DataLoader**, prefetching, and pin_memory.

PROJECTS

Computer Vision System for Camera Calibration and Augmented Reality **GitHub**

- Implemented Harris and chessboard corner detection for pose estimation using **OpenCV (C++/CUDA)**. .
- Calibrated camera; estimated pose with `findChessboardCorners`/`solvePnP`; accelerated preprocessing with **OpenCV CUDA**.
- Built a utility to generate 3D world points and project them to images; rendered axes/cubes for AR using **OpenCV** draw APIs.

Image Manipulation Tool **GitHub**

- Implemented 15+ image transforms and a desktop UI using **Java**, **Swing**, and **ImageIO**.
- Improved performance by **30%** via **MVC** refactor, cached pixel buffers, and lazy loading; applied **SOLID** design practices.
- Practiced **TDD** with **JUnit** and **Gradle**; authored 60+ unit tests, CI tasks, and generated coverage/**Javadoc** for maintainability.

Deep Learning based Spatio-Temporal Anomaly Detection in Videos **GitHub**

- Built a **3D CNN** video classifier (14 classes) in **TensorFlow/Keras**; set up frame sampling, windowed clips, and clean splits.
- Handled class imbalance with **class weights** and online **augmentation**; added **early stopping** and **ReduceLROnPlateau**.
- Processed **16,000+** surveillance clips with `tf.data` and **OpenCV**; optimized caching, prefetch, and generators.

SKILLS

Programming Languages:	C/C++, Python, JavaScript, Java, C#, TypeScript, SQL, Swift, MATLAB
Databases:	MySQL, MongoDB, PostgreSQL, SQLite, Firebase, Redis, Amazon RDS
Web Technologies:	HTML, CSS, React, Next.js, Angular, Django, Node.js, Bootstrap, REST API, Express
ML/Data Frameworks:	Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, Keras, Matplotlib

RESEARCH PUBLICATION

Covid-19 detection using chest X-rays: CNN as a classifier vs CNN as a feature extractor | **GitHub** | **Paper**